```
import java.util.*;
public class ChatBot1 {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    String msg;
    System.out.println("Chatbot: Hello! How can I help you?");
    while (true) {
      System.out.print("You: ");
      msg = sc.nextLine().toLowerCase();
      if (msg.contains("hello") || msg.contains("hi"))
        System.out.println("Chatbot: Hello! Nice to meet you.");
      else if (msg.contains("name"))
        System.out.println("Chatbot: I am a simple chatbot.");
      else if (msg.contains("how are you"))
        System.out.println("Chatbot: I am fine. Thank you!");
      else if (msg.contains("bye")){
        System.out.println("Chatbot: Goodbye!");
        break;
      } else
        System.out.println("Chatbot: Sorry, I don't understand.");}
    sc.close();
  }}
```

1. Theory

Chatbot: A chatbot is a computer program that communicates with users through text or voice.

Rule-Based Chatbot: It uses predefined IF-THEN rules to identify user input and give suitable responses.

Pattern Matching: It checks whether the user's message contains specific words or patterns.

Working:

1. User enters a message.
2. Chatbot converts it into lowercase.
3. It checks predefined patterns.
4. If a pattern matches, it displays the corresponding response.
5. If no pattern matches, it displays a default response.
6. If the user enters "bye", the chatbot stops.

2. Algorithm

1. Start.
2. Define predefined patterns and responses.
3. Display welcome message.
4. Accept user input.
5. Convert input into lowercase.
6. Check for exit words like bye, exit or quit.
7. If found, display goodbye message and stop.
8. Otherwise, check predefined patterns.
9. Display the matching response.
10. If no pattern matches, display default response.
11. Repeat until the user exits.
12. Stop.

3. Viva Questions and Answers

Q1. What is a chatbot?

A program that communicates with users.

Q2. What is a rule-based chatbot?

A chatbot that works using predefined rules.

Q3. What is pattern matching?

Checking user input for specific words or patterns.

Q4. How are rules represented?

Using IF-THEN conditions.

Q5. Why is lowercase conversion used?

To make matching case-insensitive.

Q6. What happens if no pattern matches?

It displays a default response.

Q7. What is a default response?

A reply given when no predefined rule matches.

Q8. Difference between rule-based and AI/ML chatbot?

Rule-based uses predefined rules, while AI/ML chatbots can learn from data.

Q9. What are the advantages?

- Simple to implement.
- Easy to understand.
- No training data required.

Q10. What are the limitations?

- Cannot understand complex questions.
- Cannot learn automatically.
- Gives only predefined responses.

Q11. Which Java function converts text to lowercase?

toLowerCase().

Q12. Give examples of chatbot applications.

Banking support, customer care and FAQ systems.